Engineering Chemistry

UNIT – I : Structure and Bonding Models

A clear, exam-ready walk through quantum fundamentals, the Schrödinger picture, and molecular-orbital bonding — with fully worked diagrams.

Topics in this unit

1. Fundamentals of quantum mechanics
2. Schrödinger wave equation; significance of Ψ and Ψ^2
3. Particle in a one-dimensional box
4. Molecular orbital theory (LCAO), σ & π bonds
5. Energy-level diagrams of O_2 and CO
6. π-MOs of butadiene & benzene; bond order

> * Study tip
> Every topic has a small diagram. Learn the red words first, then redraw the diagram once from memory — that is the fastest way to revise for exams.

1. Why Quantum Mechanics? The Big Picture

Classical (Newtonian) mechanics works beautifully for cricket balls and planets, but it fails for tiny particles like electrons. Electrons are so small and fast that they behave partly like particles and partly like waves. To describe them we need a new rule-book — quantum mechanics — which treats the electron as a wave and tells us only the probability of finding it somewhere.

> NOTE
> This unit builds up in a logical chain: quantum basics → the Schrödinger equation → what Ψ means → a simple solved model (particle in a box) → and finally how all this explains real chemical bonding through Molecular Orbital Theory.

2. Fundamentals of Quantum Mechanics

Three key discoveries forced scientists to abandon the purely classical picture and accept that energy and matter are quantised (come in packets).

Planck's Quantum Theory (1900)

Energy is not emitted or absorbed continuously but in small discrete packets called quanta (a packet of light = a photon). The energy of one quantum is directly proportional to the frequency of radiation.

$$E = h\nu = hc / \lambda$$

h = Planck's constant (6.626×10^{-34} J·s), ν = frequency, λ = wavelength

Photoelectric Effect — Einstein (1905)

When light of high enough frequency strikes a metal, electrons are instantly ejected. This could only be explained if light behaves as particles (photons). Each photon gives its whole energy to one electron.

$$KE_{max} = h\nu - \varphi_0$$

φ_0 = work function (minimum energy to free an electron); below the threshold frequency ν_0, no electrons come out — however bright the light.

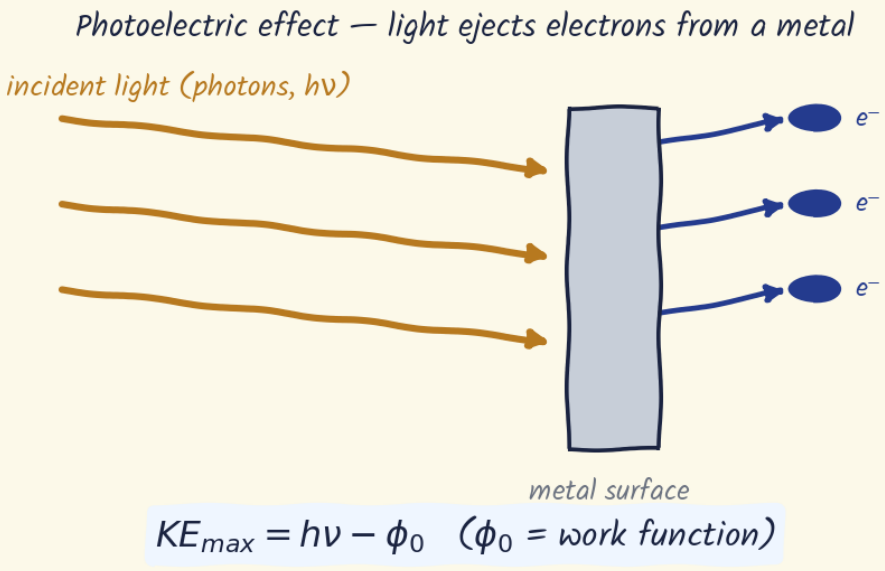


Fig 1 — Photons eject electrons from a metal surface; their kinetic energy depends on frequency, not intensity.

Wave-Particle Duality — de Broglie (1924)

If light (a wave) can behave like particles (photons), then matter (particles) can behave like waves. Every moving particle has a wavelength:

$$\lambda = h / mv = h / p$$

m = mass, v = velocity, p = momentum. Big objects → tiny λ (not noticeable); electrons → significant λ.

Heisenberg's Uncertainty Principle (1927)

It is impossible to know both the exact position and the exact momentum of an electron at the same time. The more precisely we fix one, the more uncertain the other becomes.

$$\Delta x \cdot \Delta p \geq h / 4\pi$$

Δx = uncertainty in position, Δp = uncertainty in momentum

> KEY POINT
> Because we cannot pin-point an electron's path, the old idea of fixed Bohr orbits breaks down. Instead we speak of orbitals — regions of high probability of finding the electron.

3. Schrödinger Wave Equation (1926)

Erwin Schrödinger treated the electron as a standing wave around the nucleus and wrote a wave equation for it. Solving this equation gives the allowed energy levels of the electron and the wave functions (Ψ) that describe each state. It is the heart of modern atomic structure.

Time-independent form

$$\partial^2\Psi/\partial x^2 + \partial^2\Psi/\partial y^2 + \partial^2\Psi/\partial z^2 + (8\pi^2 m/h^2)(E - V)\Psi = 0$$

The compact operator form is: $\hat{H}\,\Psi = E\,\Psi$

What each term means

- Ψ (psi) — the wave function of the electron (what we solve for).
- $\hat{H}$ — the Hamiltonian operator (represents total energy: kinetic + potential).
- E — total energy of the electron (the allowed values we get out).
- V — potential energy of the electron; m — mass; h — Planck's constant.

NOTE

Only certain values of E give acceptable (well-behaved) solutions. This is exactly why energy is quantised — it drops out of the mathematics naturally.

4. Significance of Ψ and Ψ^2

This is a favourite exam question. The wave function Ψ and its square Ψ^2 mean very different things.

DEFINITION

Ψ (wave function): the amplitude of the electron wave. It can be positive, negative or zero and has no direct physical meaning by itself. It is just a mathematical function describing the electron's state.

DEFINITION

Ψ^2 (probability density): the probability of finding the electron in a unit volume around a point. It is always positive. A region where Ψ^2 is large is exactly what we call an orbital. (Born interpretation)

Conditions for a well-behaved wave function

- *Single-valued* — one value of Ψ at each point.
- *Finite* — Ψ must not become infinite anywhere.
- *Continuous* — Ψ and its slope change smoothly.
- *Normalised* — total probability over all space = 1.

KEY POINT

Memory hook: Ψ = address, Ψ^2 = how likely the electron is at that address. Only Ψ^2 can be measured; Ψ on its own cannot.

5. Particle in a One-Dimensional Box

This is the simplest problem we can solve exactly with the Schrödinger equation, and it shows why energy is quantised in a very clear way.

The model

- A particle of mass m is trapped in a 1-D box of length a.
- Inside the box the potential energy $V = 0$ (particle moves freely).
- At and beyond the walls $V = \infty$, so the particle can never escape.

Boundary conditions & solution

Since the particle cannot exist at the walls, $\Psi = 0$ at $x = 0$ and $x = a$.

Applying these to the Schrödinger equation gives the allowed wave functions and energies:

$$\Psi_n = \sqrt{(2/a)} \cdot \sin(n\pi x/a) \quad E_n = n^2h^2 / 8ma^2$$

$n = 1, 2, 3, \ldots$ is the quantum number

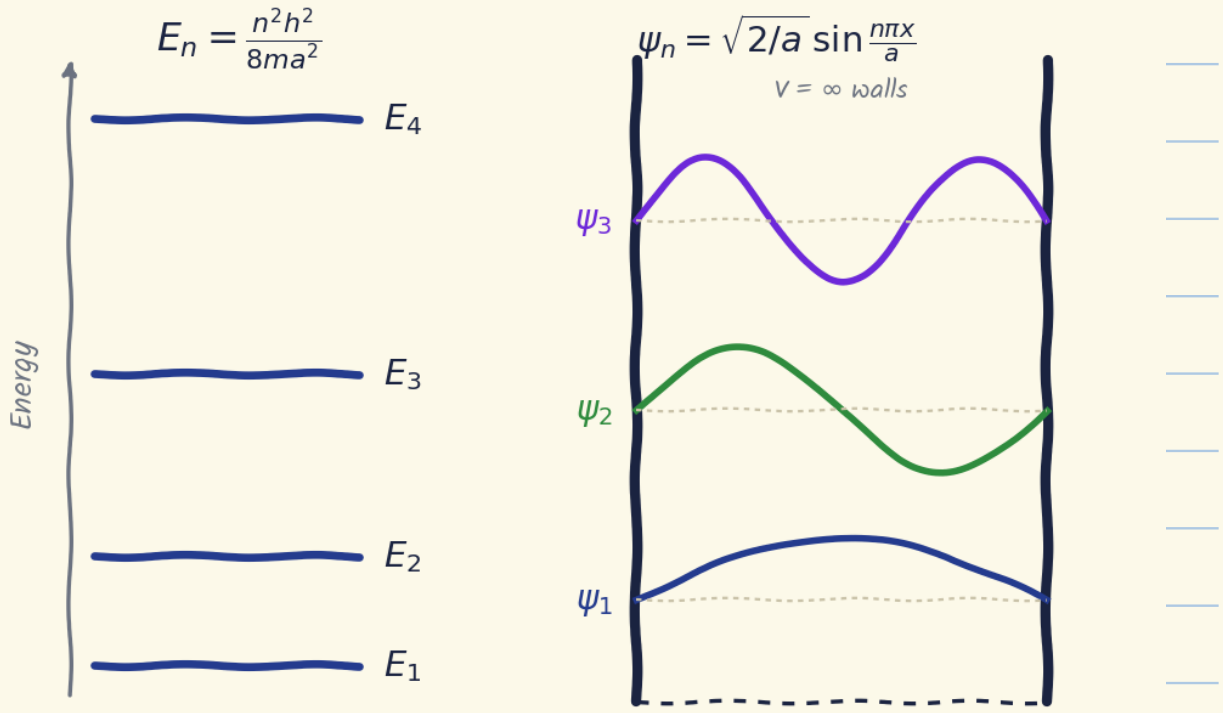


Fig 1 — Quantised energy levels (left) and the first three wave functions inside the box (right).

What this tells us

- **Energy is quantised** — only $n = 1, 2, 3\ldots$ are allowed, never values in between.
- **Zero-point energy:** the lowest level $E_1 = h^2/8ma^2$ is **not zero** — a trapped particle can never be completely at rest.
- Energy levels get **farther apart** as n increases ($E \propto n^2$): ratio $1 : 4 : 9 : 16\ldots$
- Number of nodes (points where $\psi = 0$ inside) = $n - 1$.

> **EXAMPLE**
> For $n = 1, 2, 3, 4$ the energies are in the simple ratio $1 : 4 : 9 : 16$, because $E_n \propto n^2$. So E_2 is $4\times$ E_1.

6. Molecular Orbital Theory (MOT)

Proposed by Hund and Mulliken, MOT is the most successful model of covalent bonding. Its central idea: when atoms bond, their atomic orbitals (AOs) merge and lose their individual identity, forming new molecular orbitals (MOs) that belong to the whole molecule.

LCAO — how MOs form

MOs are built by the Linear Combination of Atomic Orbitals (LCAO).

Combining n atomic orbitals always gives n molecular orbitals:

- Addition (constructive overlap, waves in phase) → bonding MO, lower energy, more stable. Example: σ, π.
- Subtraction (destructive overlap, waves out of phase) → antibonding MO (marked with *), higher energy, less stable. Example: σ*, π*.

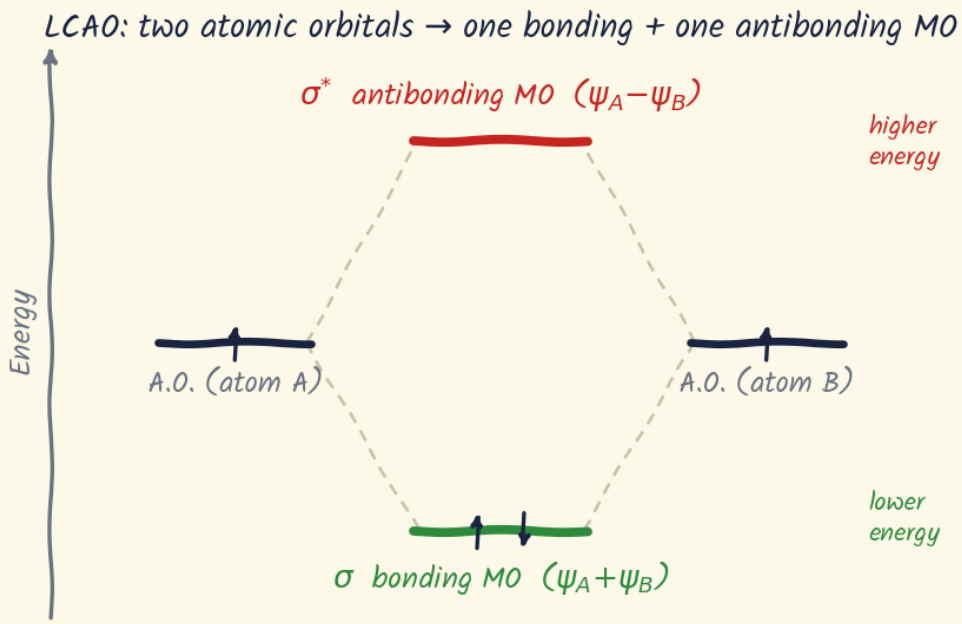


Fig 2 — Two atomic orbitals combine into one bonding (lower) and one antibonding (higher) molecular orbital.

σ and π molecular orbitals

The way orbitals overlap decides the type of MO: end-on overlap gives σ bonds (electron cloud along the bond axis) while sideways overlap gives π bonds (cloud above and below the axis).

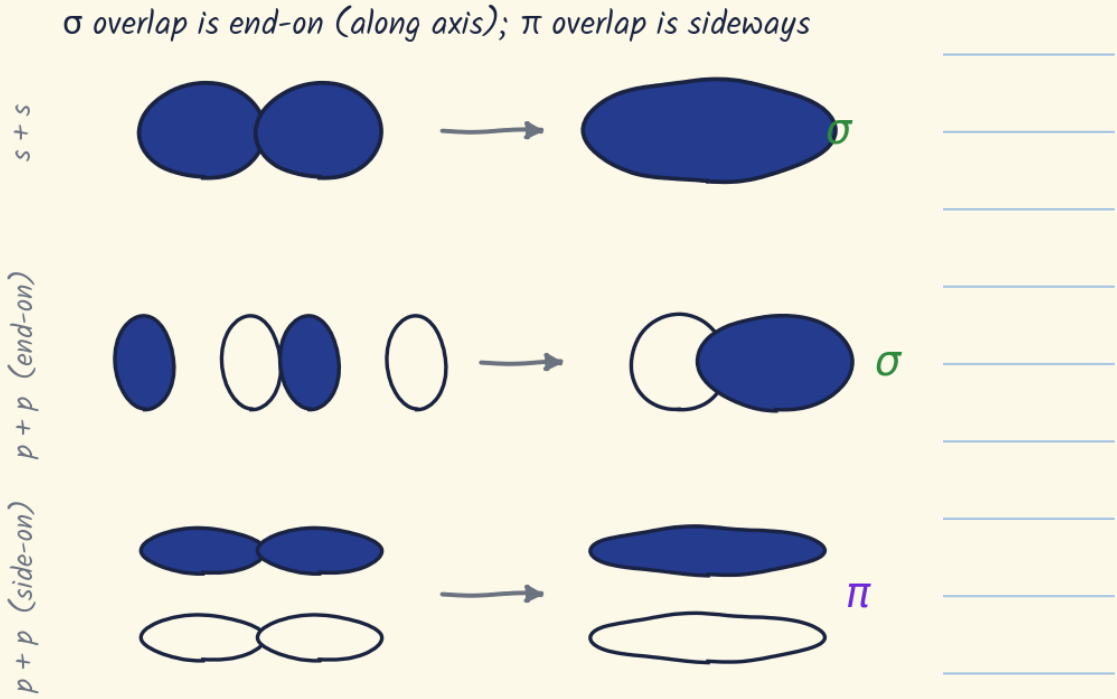


Fig 3 — σ overlap is end-on; π overlap is sideways.

Rules for filling MOs

Electrons fill molecular orbitals using the same three rules as atoms: Aufbau (lowest energy first), Pauli (max 2 e per MO, opposite spins) and Hund's rule (degenerate MOs are singly filled first).

Energy order of molecular orbitals

Up to N_2: $\sigma 1s\ \sigma^*1s\ \sigma 2s\ \sigma^*2s\ (\pi 2p_x = \pi 2p_y)\ \sigma 2p_z\ (\pi^*2p_x = \pi^*2p_y)\ \sigma^*2p_z$

For O_2, F_2 and beyond, σ2pz drops BELOW the π2p pair (no s–p mixing)

NOTE

Why two orders? For lighter molecules (B_2, C_2, N_2) s–p mixing pushes $\sigma 2p_z$ above the $\pi 2p$ orbitals. From O_2 onward this mixing is weak, so $\sigma 2p_z$ sits below $\pi 2p$. This single difference explains O_2's magnetism — see below.

7. Bond Order

Bond order is a quick number that tells us how strong and stable a bond is.

DEFINITION

Bond order (B.O.): half the difference between the number of electrons in bonding MOs (N_b) and antibonding MOs (N_a).

$$\text{Bond order} = \tfrac{1}{2}\,(N_b - N_a)$$

- B.O. > 0 → molecule is stable; B.O. = 0 → molecule does not exist.
- Higher bond order → stronger bond, shorter bond length, higher bond energy.
- Unpaired electrons → paramagnetic; all paired → diamagnetic.

8. Energy-Level Diagram of O_2

Oxygen atom ($Z = 8$): $1s^2\ 2s^2\ 2p^4$. The O_2 molecule therefore has 16 electrons, filled into MOs using the O_2-type order ($\sigma 2p_z$ below $\pi 2p$).

$$O_2:\ KK\ \sigma 2s^2\ \sigma^*2s^2\ \sigma 2p_z^{\,2}\ (\pi 2p_x^{\,2}\ \pi 2p_y^{\,2})\ (\pi^*2p_x^{\,1}\ \pi^*2p_y^{\,1})$$

KK = filled 1s MOs ($\sigma 1s^2\ \sigma^*1s^2$)

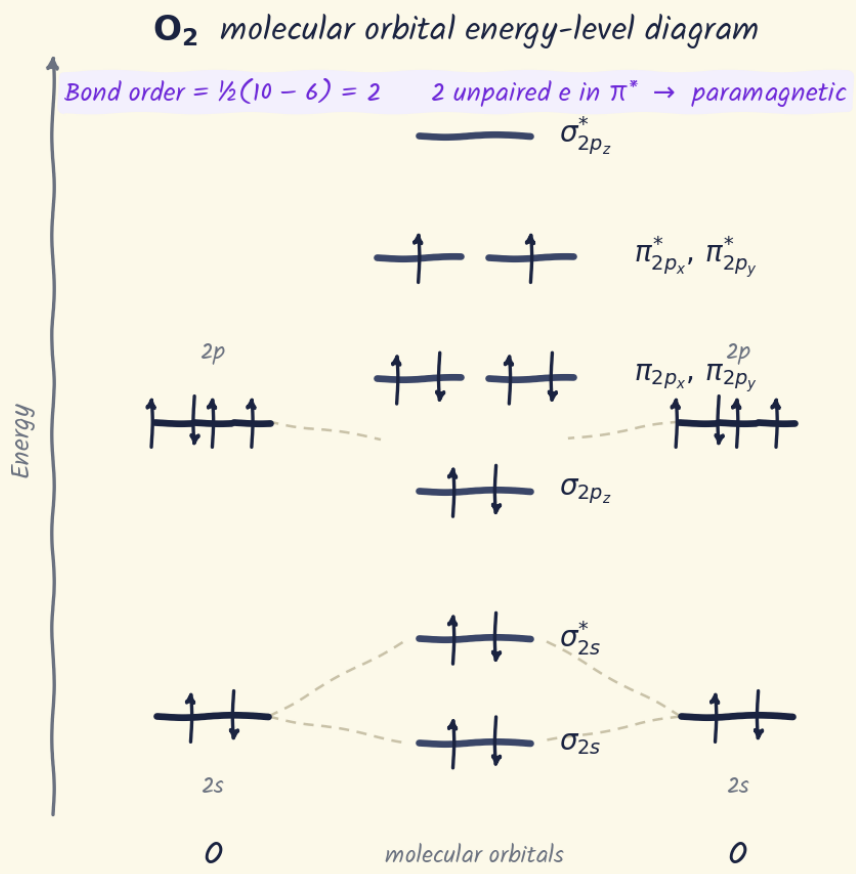


Fig 4 — Molecular-orbital energy-level diagram of O_2. The two π^* orbitals each hold one unpaired electron.

Results

- $N_b = 10$, $N_a = 6$ → Bond order = ½(10 − 6) = 2 (a double bond, O=O).
- Two unpaired electrons sit in the π^* orbitals → O_2 is paramagnetic.

> KEY POINT
>
> This is the biggest success of MOT. Lewis structures and Valence Bond Theory wrongly predict O_2 to be diamagnetic; only MOT correctly explains why liquid oxygen is attracted by a magnet.

9. Energy-Level Diagram of CO

Carbon ($Z = 6$): $1s^2\ 2s^2\ 2p^2$; Oxygen ($Z = 8$): $1s^2\ 2s^2\ 2p^4$. CO has 14 electrons and is isoelectronic with N_2, so it uses the N_2-type order ($\pi 2p$

below $\sigma 2p_z$).

$$CO:\ KK\ \sigma 2s^2\ \sigma^*2s^2\ (\pi 2p_x^{\,2}\ \pi 2p_y^{\,2})\ \sigma 2p_z^{\,2}$$

π^* and σ^*2pz remain empty

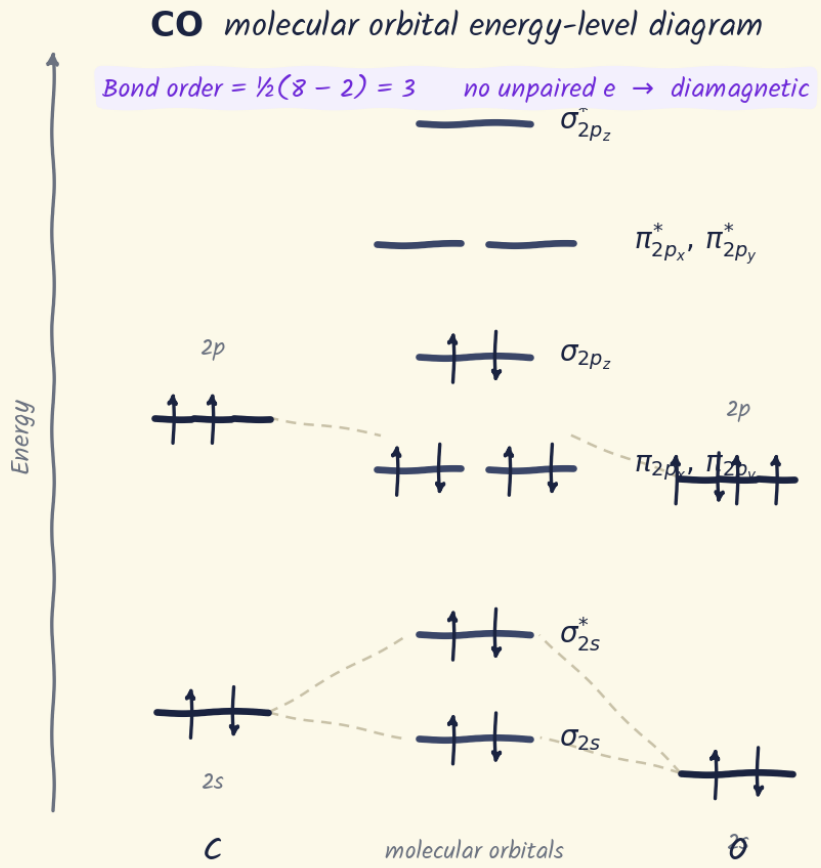


Fig 5 — MO energy-level diagram of CO (O orbitals lie a little lower as O is more electronegative).

Results

- $N_b = 10$, $N_a = 4$ → Bond order = ½(10 − 4) = 3 (a triple bond, C≡O).
- No unpaired electrons → CO is diamagnetic.
- The high bond order explains CO's very high bond energy and short bond length.

10. π-Molecular Orbitals of Butadiene

1,3-Butadiene ($CH_2=CH-CH=CH_2$) has four carbon atoms in a row, each giving one p-orbital. These 4 p-orbitals combine into 4 π molecular orbitals (ψ_1, ψ_2, ψ_3, ψ_4).

- The number of nodes increases with energy: ψ_1 has 0, ψ_2 has 1, ψ_3 has 2, ψ_4 has 3.
- There are 4 π electrons; they fill the two lowest MOs — ψ_1 and ψ_2.
- ψ_2 = HOMO (highest occupied), ψ_3 = LUMO (lowest unoccupied).

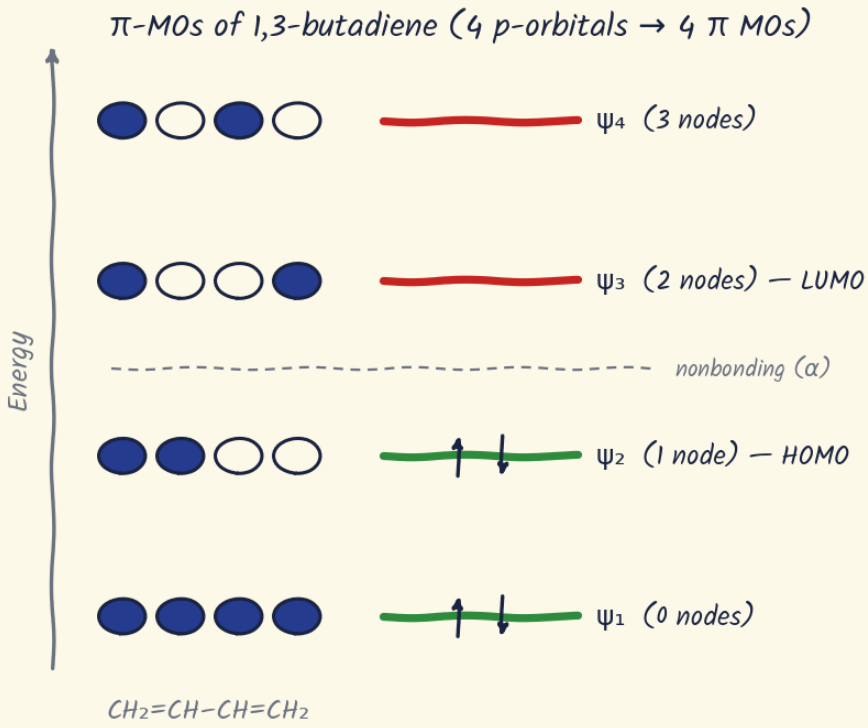


Fig 6 — The four π MOs of butadiene with their node patterns and electron filling.

> KEY POINT
> Because the π electrons are delocalised over all four carbons (not fixed between two atoms), butadiene is more stable than two isolated double bonds. This extra stability is the resonance / conjugation energy.

II. π-Molecular Orbitals of Benzene

Benzene (C_6H_6) has six carbons in a ring, each contributing a p-orbital. The 6 p-orbitals combine into 6 π MOs: three bonding (ψ_1, ψ_2, ψ_3) and three antibonding (ψ_4^*, ψ_5^*, ψ_6^*).

- ψ_1 is lowest; ψ_2 and ψ_3 are a degenerate pair (equal energy), as are ψ_4^* and ψ_5^*.
- Benzene's 6 π electrons exactly fill the 3 bonding MOs — a completely filled, very stable set.
- The result is a delocalised π cloud above and below the ring.

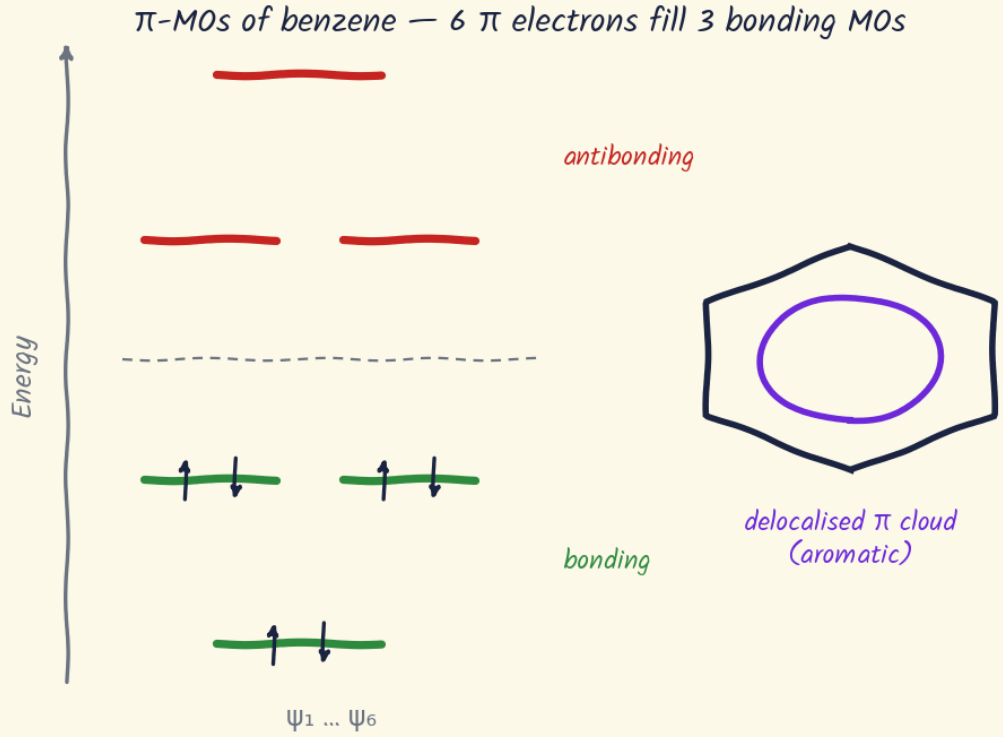


Fig 7 — Energy order of benzene's six π MOs and the delocalised π cloud over the ring.

KEY POINT

This full set of bonding MOs gives benzene its exceptional stability — its aromaticity (Hückel's rule: $4n+2$ π electrons, here $n = 1 \rightarrow 6$ electrons). This is why benzene resists addition reactions and prefers substitution.

12. Bond Orders at a Glance

Using B.O. = ½(N_b − N_a), here are common diatomic molecules. Notice how bond order tracks bond strength, and unpaired electrons decide magnetism.

Species	N_b	N_a	Bond order	Nature
H_2	2	0	1 (single)	Diamagnetic
N_2	10	4	3 (triple)	Diamagnetic
O_2	10	6	2 (double)	Paramagnetic
F_2	10	8	1 (single)	Diamagnetic
CO	10	4	3 (triple)	Diamagnetic
NO	10	5	2.5	Paramagnetic

EXAMPLE

He_2 would have N_b = 2 and N_a = 2, giving B.O. = 0. That is why the He_2 molecule does not exist — MOT predicts it cleanly.

13. Applications & Importance

- Magnetic behaviour: explains paramagnetism of O_2, B_2 and NO (unpaired e^-).
- Stability & existence: bond order tells whether a molecule/ion is stable (He_2, Be_2 with B.O. = 0 do not exist).
- Bond length & energy: higher bond order → shorter, stronger bonds.

- **Delocalisation & aromaticity:** explains the stability of butadiene, benzene and conjugated dyes.
- **Foundation for materials:** the band theory of metals, semiconductors and conductors (Unit II) grows directly out of MO ideas.

QUICK RECAP

- Quantum mechanics is needed because electrons show wave–particle duality and obey Heisenberg's uncertainty principle.
- The Schrödinger equation ($\hat{H}\Psi = E\Psi$) gives quantised energies and wave functions Ψ.
- Ψ has no physical meaning; Ψ^2 is the probability density (an orbital).
- Particle in a box: $E_n = n^2h^2/8ma^2$ — energy is quantised with a non-zero ground state (zero-point energy).
- MOT: AOs combine (LCAO) into bonding and antibonding MOs; Bond order = $\frac{1}{2}(N_b - N_a)$.
- O_2 is paramagnetic (2 unpaired e^-, B.O. 2); CO has a triple bond (B.O. 3, diamagnetic).
- Delocalised π-MOs explain the stability of butadiene and the aromaticity of benzene.

EXAM-ORIENTED QUESTIONS

Q1. State and explain Heisenberg's uncertainty principle. *[3 marks]*

Ans. It is impossible to measure the exact position and exact momentum of an electron simultaneously: $\Delta x \cdot \Delta p \geq h/4\pi$. Fixing one more precisely makes the other more uncertain — which is why electrons are described by orbitals, not fixed orbits.

Q2. What is the physical significance of Ψ and Ψ²? [3 marks]

Ans. Ψ is the wave function (amplitude of the electron wave) and has no direct physical meaning. Ψ^2 is the probability density — the chance of finding the electron in a unit volume; it is always positive and defines the orbital.

Q3. Write the Schrödinger wave equation and name its terms. [5 marks]

Ans. $\hat{H}\Psi = E\Psi$, i.e. $\partial^2\Psi/\partial x^2 + \partial^2\Psi/\partial y^2 + \partial^2\Psi/\partial z^2 + (8\pi^2 m/h^2)(E - V)\Psi = 0$, where Ψ = wave function, E = total energy, V = potential energy, m = mass, h = Planck's constant.

Q4. Discuss the particle in a 1-D box. What is zero-point energy? [5 marks]

Ans. A particle in a box of length a (V = 0 inside, ∞ at walls) has $E_n = n^2h^2/8ma^2$. Energy is quantised. The lowest level $E_1 = h^2/8ma^2$ is non-zero — this minimum energy is the zero-point energy, meaning the particle is never fully at rest.

Q5. Distinguish between bonding and antibonding molecular orbitals. [3 marks]

Ans. Bonding MOs form by constructive (in-phase) overlap, have lower energy and increase stability (σ, π). Antibonding MOs form by destructive (out-of-phase) overlap, have higher energy, carry a node between nuclei and reduce stability (σ^*, π^*).

Q6. Draw the MO diagram of O_2 and explain its paramagnetism. [5 marks]

Ans. O_2 (16 e^-): KK $\sigma 2s^2\ \sigma^*2s^2\ \sigma 2p_z^2\ \pi 2p^4\ \pi^*2p^2$. The two π^* electrons remain unpaired (Hund's rule), making O_2 paramagnetic; B.O. = ½(10−6) = 2.

Q7. Calculate the bond order of CO and give its magnetic nature. [3 marks]

Ans. CO (14 e^-, N_2-type order): $N_b = 10$, $N_a = 4$, so B.O. = ½(10−4) = 3 (triple bond). All electrons are paired → diamagnetic.

Q8. Why does the He_2 molecule not exist? [2 marks]

Ans. In He_2, $N_b = 2$ and $N_a = 2$, giving bond order 0. With no net bonding, He_2 is not a stable molecule.

Q9. Explain the π-MOs and aromaticity of benzene. [3 marks]

Ans. Benzene's 6 p-orbitals give 6 π MOs (3 bonding, 3 antibonding). Its 6 π electrons fully fill the 3 bonding MOs, creating a stable delocalised π cloud — this is aromaticity (Hückel 4n+2 rule, n = 1).

Q10. State the de Broglie relation and its significance. [2 marks]

Ans. $\lambda = h/mv$. Every moving particle has an associated wavelength; it is significant for light, fast particles like electrons and is the basis of their wave behaviour.

Q11. Define an orbital. How is it different from a Bohr orbit? [3 marks]

Ans. An orbital is a region of space where the probability (Ψ^2) of finding an electron is high. Unlike a fixed Bohr orbit (a definite circular path), an orbital gives only a probability, which agrees with Heisenberg's uncertainty principle.

Q12. What is LCAO? [2 marks]

Ans. Linear Combination of Atomic Orbitals — atomic orbitals are added and subtracted to build molecular orbitals. Combining n atomic orbitals always gives n molecular orbitals (bonding + antibonding).

Q13. Why does the N_2-type MO order differ from the O_2-type order? [3 marks]

Ans. In lighter molecules (up to N_2), s–p mixing pushes $\sigma 2p_z$ above the $\pi 2p$ pair. From O_2 onward this mixing is weak, so $\sigma 2p_z$ lies below $\pi 2p$. This is why O_2 ends up with unpaired π^* electrons (paramagnetic).

Q14. Compare σ and π bonds. [3 marks]

Ans. A σ bond forms by end-on (axial) overlap and is stronger, with electron density along the bond axis. A π bond forms by sideways overlap of p-orbitals and is weaker, with electron density above and below the axis.

Q15. Calculate the bond order of N_2 and give its magnetic nature. [3 marks]

Ans. N_2 (14 electrons): $N_b = 10$, $N_a = 4$, so B.O. = $\frac{1}{2}(10 - 4) = 3$ (a triple bond). All electrons are paired, so N_2 is diamagnetic.

Q16. State Planck's quantum theory. [2 marks]

Ans. Energy is emitted or absorbed only in small discrete packets called quanta, with $E = h\nu$ (proportional to frequency) — not continuously.

Q17. What is a node in a wave function? [2 marks]

Ans. A node is a point or region where $\Psi = 0$, so the probability of finding the electron there is zero. For a particle in a box the number of nodes = $n - 1$.